

Sample Test 1 — Solutions

Question 1

Let

$$\phi = \exists z (P(x, z) \wedge (\forall y (\neg Q(y, x) \vee P(y, t)))) \quad \text{and} \quad \psi = \forall z P(y, z) \rightarrow (\forall x \forall y (\neg Q(y, x) \vee R(y, z))).$$

(a) Parse trees and free/bound variables

Parse tree for ϕ :

$$\exists z \left(P(x, z) \wedge \forall y (\neg Q(y, x) \vee P(y, t)) \right)$$

- z is **bound** (bound by $\exists z$).
- y is **bound** (bound by $\forall y$).
- x is **free**.
- t is **free**.

Parse tree for ψ :

$$\forall z P(y, z) \rightarrow \forall x \forall y (\neg Q(y, x) \vee R(y, z))$$

Note: z appears both bound (on the left under $\forall z$) and free (on the right inside $R(y, z)$, outside the scope of $\forall z$).

- z on the left: **bound** (bound by $\forall z$).
- z on the right in $R(y, z)$: **free**.
- x on the right: **bound** (bound by $\forall x$).
- y on the left in $P(y, z)$: **free**.
- y on the right: **bound** (bound by $\forall y$).

(b) Substitutions

Compute $\phi[f(y)/x]$: Replace the free occurrence of x with $f(y)$.

$$\phi[f(y)/x] = \exists z (P(f(y), z) \wedge \forall y (\neg Q(y, f(y)) \vee P(y, t))).$$

Compute $\psi[z/y]$: Replace the free occurrence of y (in $P(y, z)$ on the left) with z . The right-hand side y is bound and is not substituted.

$$\psi[z/y] = \forall z P(z, z) \rightarrow (\forall x \forall y (\neg Q(y, x) \vee R(y, z))).$$

Question 2

(a) $\exists x(P(x) \wedge Q(x)), \forall x(Q(x) \rightarrow R(x)) \vdash \exists x(P(x) \wedge R(x))$

1.	$\exists x(P(x) \wedge Q(x))$	premise
2.	$\forall x(Q(x) \rightarrow R(x))$	premise
3.	$x_0, P(x_0) \wedge Q(x_0)$	assumption from $\exists E$ (1)
4.	$P(x_0)$	$\wedge_1 E$ (3)
5.	$Q(x_0)$	$\wedge_2 E$ (3)
6.	$Q(x_0) \rightarrow R(x_0)$	$\forall E$ (2)
7.	$R(x_0)$	$\rightarrow E$ (6,5)
8.	$P(x_0) \wedge R(x_0)$	$\wedge I$ (4,7)
9.	$\exists x(P(x) \wedge R(x))$	$\exists I$ (8)
10.	$\exists x(P(x) \wedge R(x))$	$\exists E$ (1, 3–9)

(b) $\exists x P(x), \forall x \forall y (P(x) \rightarrow Q(y)) \vdash \forall y Q(y)$

1.	$\exists x P(x)$	premise
2.	$\forall x \forall y (P(x) \rightarrow Q(y))$	premise
3.	$x_0, P(x_0)$	assumption from $\exists E$ (1)
4.	y_0	arbitrary
5.	$\forall y (P(x_0) \rightarrow Q(y))$	$\forall E$ (2)
6.	$P(x_0) \rightarrow Q(y_0)$	$\forall E$ (5)
7.	$Q(y_0)$	$\rightarrow E$ (6,3)
8.	$\forall y Q(y)$	$\forall I$ (4–7)
9.	$\forall y Q(y)$	$\forall E$ (1, 3–8)

Question 3

(a) $\forall x (P(x) \rightarrow Q(f(x)))$

Model $\mathcal{M}_1$ (Satisfies):

- Domain $\mathcal{D} = \{1, 2\}$
- $f(n) = n$ (identity function)
- $P = \{1\}, Q = \{1, 2\}$

Explanation: For $x = 1$: $P(1) \rightarrow Q(f(1)) = T \rightarrow T = T$. For $x = 2$: $P(2) \rightarrow Q(f(2)) = F \rightarrow T = T$. All cases are true.

Model $\mathcal{M}_2$ (Does not satisfy):

- Domain $\mathcal{D} = \{1\}$
- $f(1) = 1$
- $P = \{1\}, Q = \emptyset$

Explanation: For $x = 1$: $P(1) \rightarrow Q(f(1)) = T \rightarrow F = F$. The universal is falsified.

(b) $\exists x \forall y ((P(x, y) \wedge P(y, x)) \vee (Q(x, y) \wedge R(y, x)))$

Model $\mathcal{M}_1$ (Satisfies):

- Domain $\mathcal{D} = \{1, 2\}$
- $P = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$ (true for all pairs), $Q = \emptyset$, $R = \emptyset$

Explanation: Take $x = 1$. For any $y \in \{1, 2\}$: $P(1, y) \wedge P(y, 1) = T \wedge T = T$. So the disjunction holds for all y , satisfying the formula.

Model $\mathcal{M}_2$ (Does not satisfy):

- Domain $\mathcal{D} = \{1, 2\}$
- $P = \emptyset$, $Q = \emptyset$, $R = \emptyset$

Explanation: For any choice of x , both $P(x, y) \wedge P(y, x)$ and $Q(x, y) \wedge R(y, x)$ are false for all y . The disjunction is always false, so no witness x exists.

Question 4

(a) $\exists x P(x) \wedge \exists x Q(x) \models \exists x (P(x) \wedge Q(x))$? **Does not hold.**

Counterexample: Let $\mathcal{M}$ be a model with:

- Domain $\mathcal{D} = \{1, 2\}$
- $P = \{1\}$, $Q = \{2\}$

Explanation: $\exists x P(x)$ is true (witnessed by $x = 1$) and $\exists x Q(x)$ is true (witnessed by $x = 2$), so the premise holds. However, $\exists x (P(x) \wedge Q(x))$ requires a single x satisfying both: $P(1) \wedge Q(1) = T \wedge F = F$ and $P(2) \wedge Q(2) = F \wedge T = F$. So the conclusion is false. The entailment does not hold.

(b) $\exists y \forall x P(x, y) \models \forall x \exists y P(x, y)$? **Holds.**

Let $\mathcal{M}$ be any model such that $\mathcal{M} \models \exists y \forall x P(x, y)$. Then there exists some $y_0 \in \mathcal{D}$ such that $P(x, y_0)$ holds for every $x \in \mathcal{D}$. Now take any arbitrary $x_0 \in \mathcal{D}$. Since $P(x_0, y_0)$ holds, y_0 witnesses $\exists y P(x_0, y)$. Since x_0 was arbitrary, $\forall x \exists y P(x, y)$ holds. Therefore the entailment holds.

Question 5

(a) $\phi = ((p \rightarrow q) \wedge p) \vee q$

To check validity, we negate and construct the tableau for $\neg\phi$:

$$\neg(((p \rightarrow q) \wedge p) \vee q) \equiv \neg((p \rightarrow q) \wedge p) \wedge \neg q \equiv (\neg(p \rightarrow q) \vee \neg p) \wedge \neg q$$

Applying the $\wedge$ -rule:

$$\phi_1 = \{(\neg(p \rightarrow q) \vee \neg p), \neg q\}$$

Since $\neg(p \rightarrow q) \equiv p \wedge \neg q$, expand:

$$\phi_2 = \phi_1 \cup \{p \wedge \neg q\} \cup \{\neg q\}$$

Now apply the $\vee$ -rule to $(\neg(p \rightarrow q) \vee \neg p)$, branching into:

Branch 1: $\neg(p \rightarrow q)$, i.e. $p \wedge \neg q$. Combined with $\neg q$: $\{p, \neg q, \neg q\}$ — consistent, no immediate contradiction.

But now check $\phi = ((p \rightarrow q) \wedge p) \vee q$ under $p = T$, $q = F$: $(p \rightarrow q) = F$, so $(F \wedge T) \vee F = F$. Branch 1 is open — ϕ is **false** here.

Wait — re-examine ϕ directly. Under $p = T, q = F$:

$$((T \rightarrow F) \wedge T) \vee F = (F \wedge T) \vee F = F \vee F = F.$$

So ϕ is **not valid**. To check satisfiability, under $p = F, q = F$:

$$((F \rightarrow F) \wedge F) \vee F = (T \wedge F) \vee F = F.$$

Under $p = T, q = T$:

$$((T \rightarrow T) \wedge T) \vee T = (T \wedge T) \vee T = T.$$

So ϕ is **satisfiable** (true when $q = T$) but **not valid**.

(b) $\psi = \forall x \neg P(x) \vee \exists x (P(x) \vee Q(x))$

To check validity, negate and construct the tableau for $\neg\psi$:

$$\begin{aligned} \neg\psi &= \neg(\forall x \neg P(x) \vee \exists x (P(x) \vee Q(x))) \equiv \neg\forall x \neg P(x) \wedge \neg\exists x (P(x) \vee Q(x)) \\ &\equiv \exists x P(x) \wedge \forall x \neg(P(x) \vee Q(x)) \equiv \exists x P(x) \wedge \forall x (\neg P(x) \wedge \neg Q(x)) \end{aligned}$$

Apply the $\wedge$ -rule:

$$\psi_1 = \{\exists x P(x), \forall x (\neg P(x) \wedge \neg Q(x))\}$$

Apply $\exists$ -rule: introduce fresh constant a with $P(a)$.

$$\psi_2 = \psi_1 \cup \{P(a)\}$$

Apply $\forall$ -rule with $x = a$: $\neg P(a) \wedge \neg Q(a)$, giving $\neg P(a)$.

$$\psi_3 = \psi_2 \cup \{\neg P(a)\}$$

Now $\{P(a), \neg P(a)\}$ is a contradiction. The branch **closes**. Since all branches close, $\neg\psi$ is unsatisfiable. Therefore ψ is **valid**.